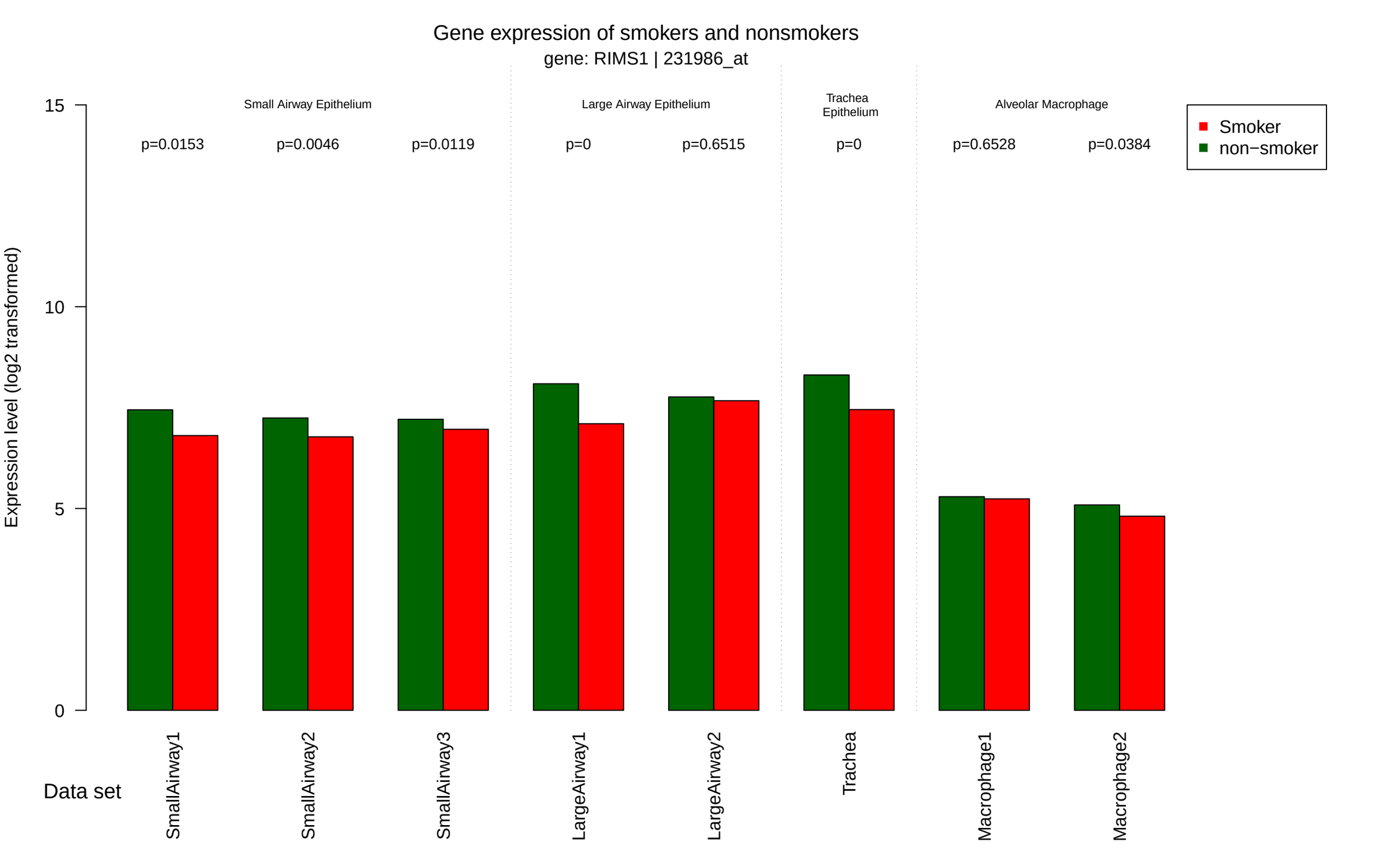
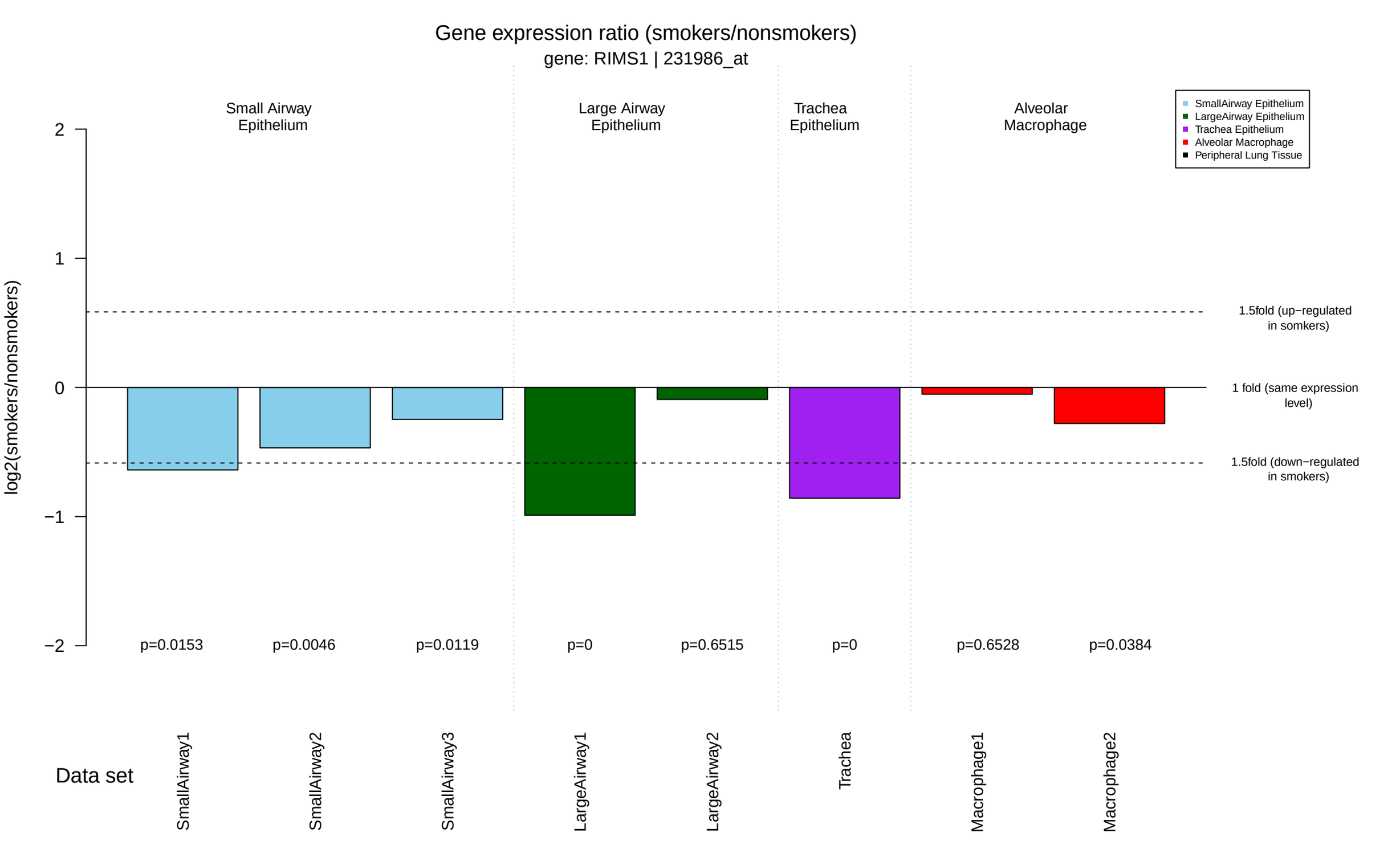
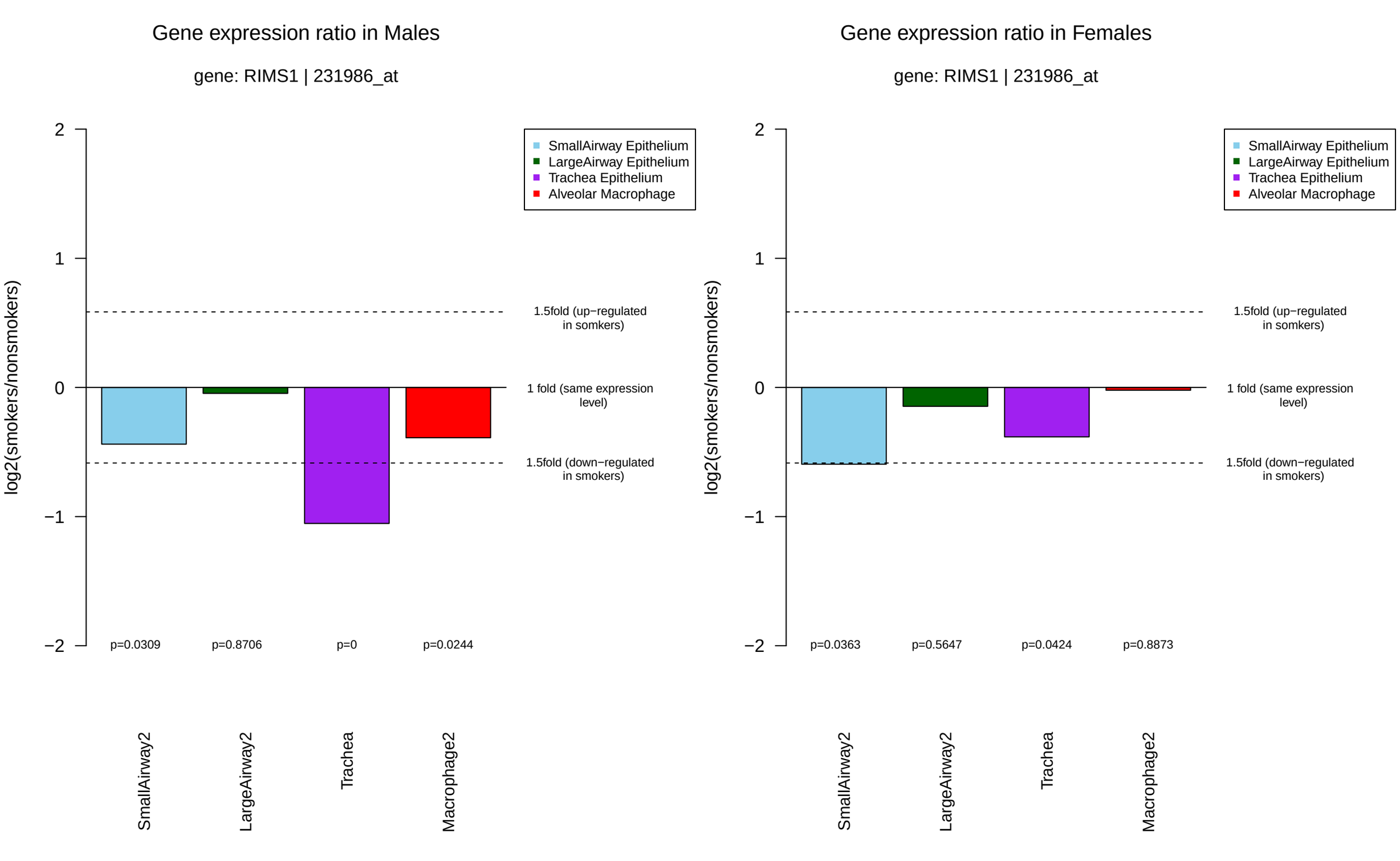
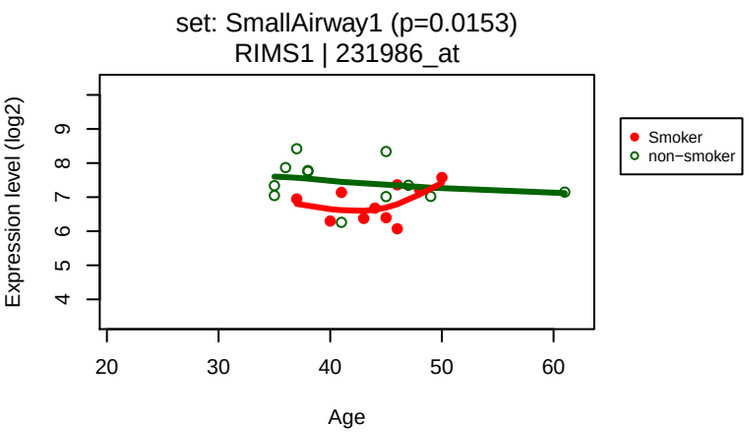
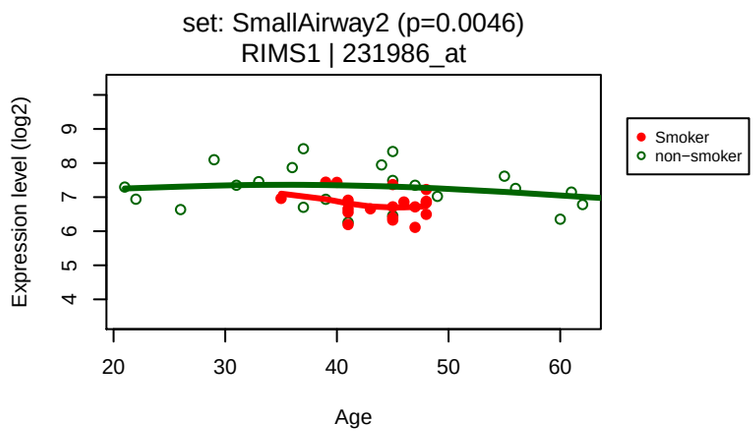
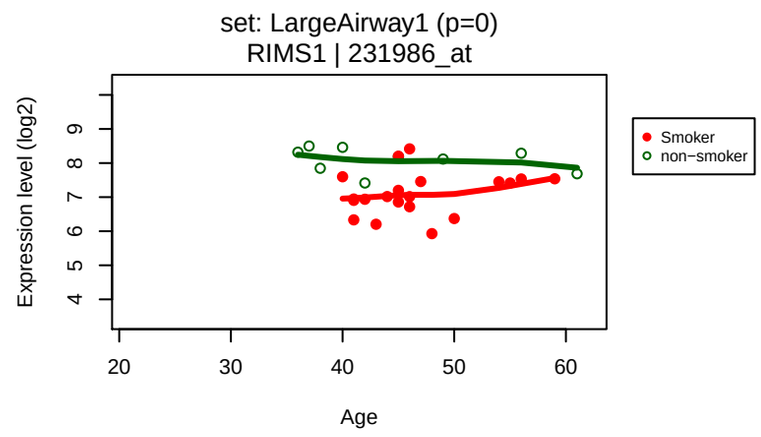

Gene expression of smokers and nonsmokers
gene: RIMS1 | 231986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0153
p=0.0046
p=0.0119
p=0
p=0.6515
p=0
p=0.6528
p=0.0384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

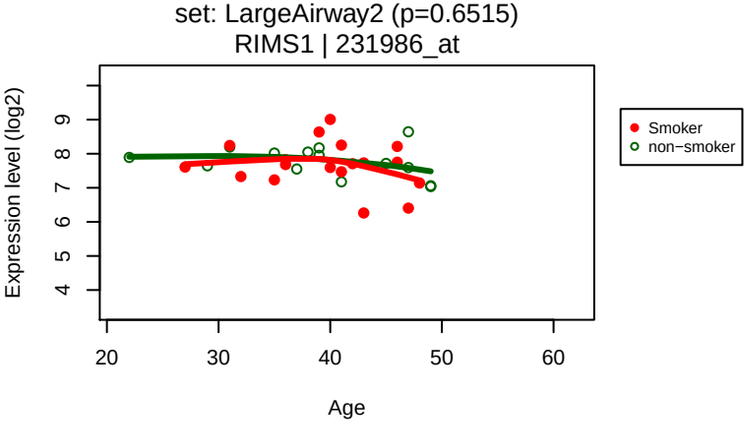

Gene expression ratio (smokers/nonsmokers)
gene: RIMS1 | 231986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0153
p=0.0046
p=0.0119
p=0
p=0.6515
p=0
p=0.6528
p=0.0384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

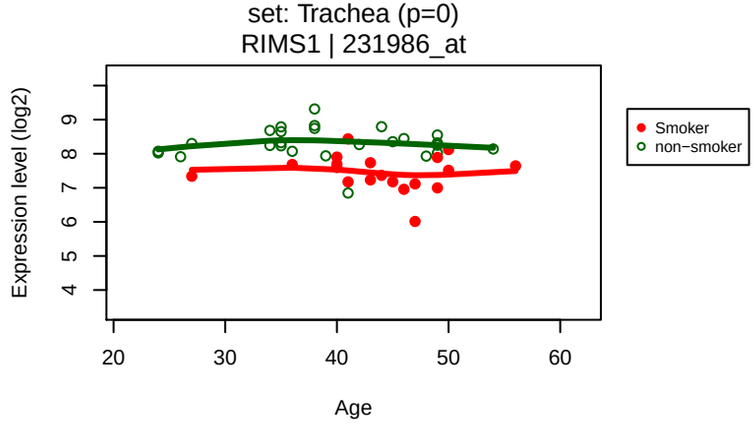

Gene expression ratio in Males
gene: RIMS1 | 231986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0309
p=0.8706
p=0
p=0.0244
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIMS1 | 231986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0363
p=0.5647
p=0.0424
p=0.8873
SmallAirway2
LargeAirway2
Trachea
Macrophage2

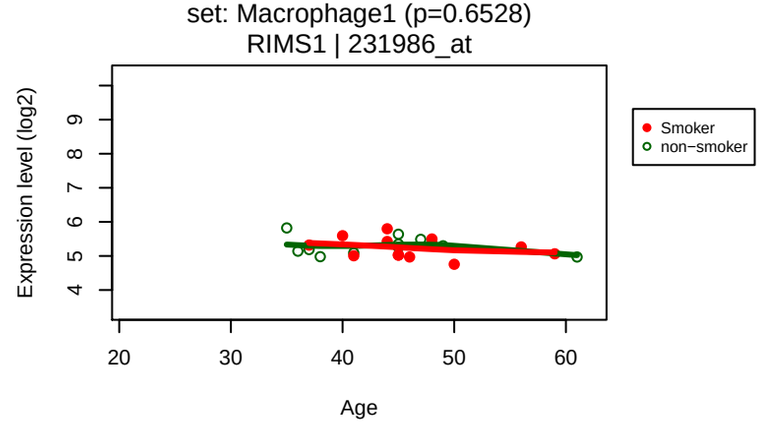

set: SmallAirway1 (p=0.0153)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker

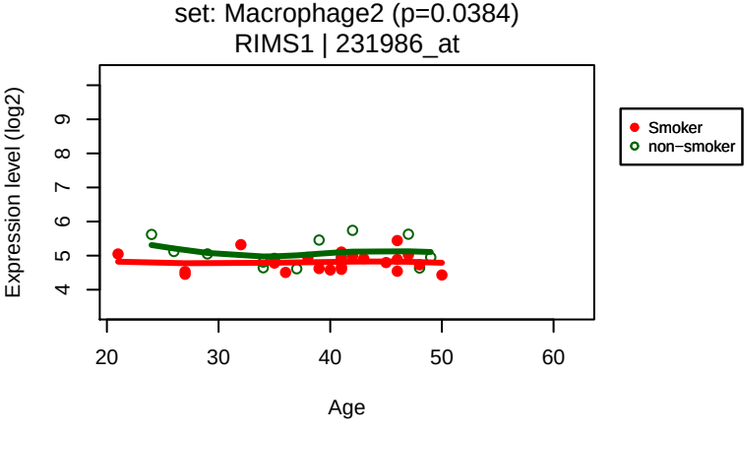

set: SmallAirway2 (p=0.0046)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker

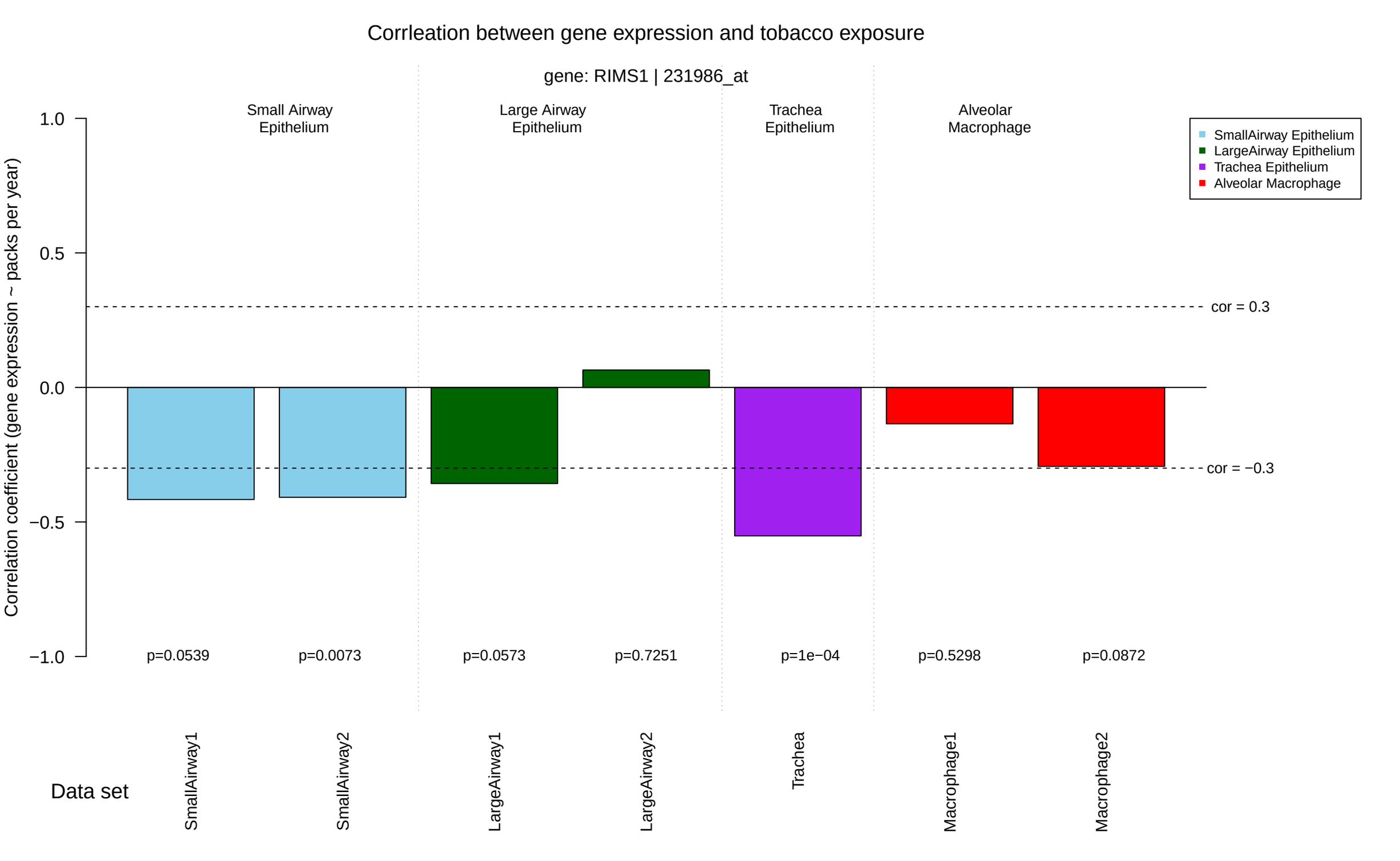

set: LargeAirway1 (p=0)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6515)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6528)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0384)
RIMS1 | 231986_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIMS1 | 231986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0539
p=0.0073
p=0.0573
p=0.7251
p=1e−04
p=0.5298
p=0.0872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2